Pricing and hedging of financial derivatives — Peter Tankov
2024–2025, main session

No documents/calculators allowed, except for 1 sheet of paper (A4, two-sided) with handwritten notes. Exam length 2 hours.

If you have difficulties with a question, assume that its result is given and move on.

Exercise 1: Log-normal mixture model

In this exercise we study a variant of the local volatility model, developed in [Brigo, D., and Mercurio, F. A mixed-up smile. Risk September 2000, 123-126].

Fix a positive integer $N \geq 1$, Let W be a standard Brownian motion, $r > 0$ be the interest rate and $\sigma_1, \dots, \sigma_N$ be positive constants. Consider N processes with risk-neutral dynamics given by

$$\frac{dS_t^i}{S_t^i} = rdt + \sigma_i dW_t, \quad S_0^i = S_0.$$

in other words, for each i, S^i follows the standard Black-Scholes model with volatility σ_i and with initial value S_0 (common for all processes).

1. Recall the formula for the price of the call option on S_T^i with strike K, denoted by $C^i(T, K)$.
2. Recall (or derive) the formulas for the derivatives $\frac{\partial C^i(T,K)}{\partial K}$, $\frac{\partial^2 C^i(T,K)}{\partial K^2}$ and $\frac{\partial C^i(T,K)}{\partial T}$. Check your result by applying Dupire's formula to C^i.

We now consider an option price surface $C(T, K)$, $T \geq 0$, $K \geq 0$, given by the weighted sum of Black-Scholes prices with volatilities $\sigma_1, \dots, \sigma_N$:

$$C(T, K) = \sum_{i=1}^{N} w_i C^i(T, K), \tag{1}$$

where $w_1, \dots, w_N$ are positive constant weights such that $w_1 + \dots + w_N = 1$.

3. Recall the five conditions of arbitrage-free option price surfaces (from the course). Check that these conditions are satisfied by the surface (1).
4. Using Dupire's formula, compute the local volatility corresponding to option price function $C(T, K)$. Show that the local volatility may be represented as follows:

$$\sigma^2(T, K) = \sum_{i=1}^{N} \sigma_i^2 \lambda_i(T, K),$$

where $\lambda_i(T, K)$, $i = 1, \dots, N$ are positive weights summing up to one, to be determined. Deduce that the local volatility satisfies $\min_i \sigma_i \leq \sigma(T, K) \leq \max_i \sigma_i$ for all T, K.

From now on, we consider the risk-neutral dynamics of the underlying

$$\frac{dS_t}{S_t} = rdt + \sigma(t, S_t) dW_t.$$

This model will be referred to as log-normal mixture model.

5. Describe the hedging strategy for a call option with maturity T and strike K in the log-normal mixture model.

6. Based on the result of question 4, and the robustness of the Black-Scholes formula, give a sub-hedging and a super-hedging strategy for the call option in the log-normal mixture model using the Black-Scholes delta.

7. Recall the formula for the model-free price of a variance swap in terms of European call and put prices (more precisely, we are interested in the price of the variable leg of the swap, $\frac{1}{T}\int_0^T \sigma_s^2 ds$, as computed in the course). What is the price of a variance swap in the Black-Scholes model with constant volatility?

8. From the option price formula (1) and the variance swap price in the Black-Scholes model, deduce the price of the variance swap in the log-normal mixture model.

Exercise 2: Implied volatility in the log-normal mixture model

We use the same notation and the same assumptions as in Exercise 1.

1. Recall the definition of the implied volatility. From the representation (1), deduce that the implied volatility in the log-normal mixture model, denoted by $I(T, K)$, satisfies

$$\min_i \sigma_i \leq I(T, K) \leq \max_i \sigma_i$$

2. Recall that at the money forward (ATMF) implied volatility $I(T, S_0e^{rT})$ satisfies

$$I(T, S_0e^{rT}) \sim \frac{C(T, S_0e^{rT})}{S_0}\sqrt{\frac{2\pi}{T}} \quad \text{as} \quad T \to 0.$$

   Use this formula to compute the short-term ATMF volatility in the log-normal mixture model.

3. Recall (or derive) the formula for the derivative of the Black-Scholes option price with respect to the volatility $\frac{\partial C^{BS}(T,K,\sigma)}{\partial \sigma}$. Compute also the second derivative $\frac{\partial^2 C^{BS}(T,K,\sigma)}{\partial \sigma^2}$ and show that

   (a) The ATMF call option price (for $K = S_0e^{-rT}$) in the Black-Scholes model is concave as function of the volatility σ and

   (b) When the strike is sufficiently large or sufficiently small, the call option price in the Black-Scholes model is convex as function of the volatility σ for $\sigma \leq \max_i \sigma_i$.

4. Using the representation (1), Jensen's inequality, and previous question, show that

   (a) The ATMF implied volatility in the log-normal mixture model satisfies $I(T, S_0e^{rT}) \leq \sum_{i=1}^N w_i\sigma_i$.

   (b) When the strike is sufficiently large or sufficiently small, the implied volatility in the log-normal mixture model satisfies $I(T, K) \geq \sum_{i=1}^N w_i\sigma_i$.

5. In the standard Black-Scholes model, define a new probability Q^* by

$$\frac{dQ^*}{dQ}|_{\mathcal{F}_t} = \frac{S_te^{-rt}}{S_0}.$$

   Show that $S_t^* := \frac{S_0^2e^{2rt}}{S_t}$ under Q^* has the same law as S under Q. Deduce that

$$C^{BS}(T, K) = \frac{Ke^{-rT}}{S_0}P^{BS}\left(T, \frac{S_0^2e^{2rT}}{K}\right),$$

   where P^{BS} denotes the put option price in the Black-Scholes model.

6. Use the result of the previous question to show that the implied volatility smile in the log-normal mixture model possesses the following symmetry property:

$$I(T, K) = I\left(T, \frac{S_0^2e^{2rT}}{K}\right).$$

Solution

Exercise 1

1. By Black-Scholes formula,

$$C^i(T,K) = S_0 N(d_1^i) - Ke^{-rT}N(d_2^i), \quad d_{12}^i = \frac{\log(S_0/Ke^{-rT}) \pm \frac{1}{2}\sigma_i^2 T}{\sigma_i\sqrt{T}}$$

2. Differentiating the above expression, we get:

$$\begin{aligned}
\frac{\partial C^i(T,K)}{\partial K} &= -e^{-rT}N(d_2^i) \\
\frac{\partial^2 C^i(T,K)}{\partial K^2} &= \frac{e^{-rT}n(d_2^i)}{K\sigma_i\sqrt{T}} \\
\frac{\partial C^i(T,K)}{\partial T} &= rKe^{-rT}N(d_2^i) + Ke^{-rT}n(d_2^i)\frac{\sigma_i}{2\sqrt{T}}.
\end{aligned}$$

   Substituting these derivatives into Dupire's formula, we obtain

$$2\frac{\frac{\partial C^i(T,K)}{\partial T} + rK\frac{\partial C^i(T,K)}{\partial K}}{K^2\frac{\partial^2 C^i(T,K)}{\partial K^2}} = \sigma_i^2$$

   as expected.

3. The following no arbitrage conditions have been studied in the lectures:

   (a) Convexity in K: for all $T \geq 0$, $C(T,\cdot)$ is convex.

   (b) Monotony in T: for all $K \geq 0$, $C(K,\cdot)$ is increasing.

   (c) Bounds: $(S-K)^+ \leq C(T,K) \leq S \quad \forall K \geq 0, \forall T \geq 0$.

   (d) Value at expiry: $C(0,K) = (S-K)^+ \quad \forall K > 0$.

   (e) Large strike limit: $\lim_{K\to\infty} C(T,K) = 0 \quad \forall T \geq 0$.

   All these conditions for the price (1) easily follow from the same conditions for the Black-Scholes price (which is arbitrage free because it is given by risk-neutral expectation).

4. By Dupire's formula,

$$\sigma^2(T,K) = 2\frac{\sum_{i=1}^N w_i\left(\frac{\partial C^i(T,K)}{\partial T} + rK\frac{\partial C^i(T,K)}{\partial K}\right)}{\sum_{i=1}^N w_i\left(K^2\frac{\partial^2 C^i(T,K)}{\partial K^2}\right)} = \frac{\sum_{i=1}^N w_i\sigma_i n(d_2^i)}{\sum_{i=1}^N w_i\sigma_i^{-1}n(d_2^i)}$$

   Letting $\lambda_i(T,K) := \frac{w_i\sigma_i^{-1}n(d_2^i)}{\sum_{i=1}^N w_i\sigma_i^{-1}n(d_2^i)}$, we get the desired representation, from which the bounds follow easily.

5. The hedging strategy is the standard hedging strategy for the local volatility model studied during the lectures: the option price is given by

$$C(t,S,T,K) = \mathbb{E}[e^{-r(T-t)}(S_T-K)^+|S_t = S]$$

   and the quantity of risky asset in the portfolio is $\delta_t = (\partial C/\partial S)(t,S_t)$. Note that this is not the Black-Scholes delta.

6. Because the local volatility is bounded from above by $\max_i \sigma_i$ and from below by $\min_i \sigma_i$, by the robustness of the Black-Scholes formula, we know that a call option can be super-hedged by the Black-Scholes delta hedging strategy with volatility $\max_i \sigma_i$ and sub-hedged by the Black-Scholes delta hedging strategy with volatility $\min_i \sigma_i$.

7. From the course, we know that the model-free price of a variance swap is given by

$$\frac{2}{T}\int_0^F \frac{P(T,K)}{K^2}dK + \frac{2}{T}\int_F^\infty \frac{C(T,K)}{K^2}dK.$$

Applying this formula in the Black-Scholes model, we get:

$$\frac{2}{T}\int_0^F \frac{P^i(T,K)}{K^2}dK + \frac{2}{T}\int_F^\infty \frac{C^i(T,K)}{K^2}dK = \sigma_i^2.$$

8. Applying the above formula in the log-normal mixture model with formula (1), we get that the price of the variance swap equals

$$\sum_{i=1}^N w_i \left(\frac{2}{T}\int_0^F \frac{P^i(T,K)}{K^2}dK + \frac{2}{T}\int_F^\infty \frac{C^i(T,K)}{K^2}dK\right) = \sum_{i=1}^N w_i\sigma_i^2.$$

Exercise 2

1. The implied volatility is the solution of

$$C^{BS}(T,K,I(T,K)) = \sum_{i=1}^N w_i C^i(T,K).$$

Since the Black-Scholes price is increasing in volatility and the weights sum up to one,

$$C^{BS}(T,K,\min_i \sigma_i) \le \sum_{i=1}^N w_i C^i(T,K) \le C^{BS}(T,K,\max_i \sigma_i),$$

from which the bounds follow.

2. Applying the asymptotic formula first to C^i, we get that $C^i(T, S_0e^{rT}) \sim S_0\sigma_i\sqrt{\frac{T}{2\pi}}$. Applying it now to the log-normal mixture model, we get:

$$I(T,S_0e^{rT}) \sim \frac{\sum_{i=1}^N w_i C^i(T,S_0e^{rT})}{S_0}\sqrt{\frac{2\pi}{T}} \sim \sum_{i=1}^N w_i\sigma_i.$$

3. Recall the expression for the Black-Scholes vega and its derivative (the vomma):

$$\frac{\partial C^{BS}}{\partial\sigma} = Sn(d_1)\sqrt{T}, \qquad \frac{\partial^2 C^{BS}}{\partial\sigma^2} = n(d_1)\frac{d_1d_2\sqrt{T}}{\sigma} = n(d_1)\frac{\sqrt{T}}{\sigma}\frac{\log^2(S_0/Ke^{-rT}) - \sigma^4T^2/4}{\sigma^2 T}$$

The required concavity and convexity properties are easily deduced from this formula.

4. For the ATMF implied volatility, since the Black-Scholes price is concave in σ, we get by Jensen's inequality:

$$\sum_{i=1}^N w_i C^i(T,K) \le C^{BS}(T,K,\sum_{i=1}^N w_i\sigma_i),$$

which shows that the implied volatility is bounded from above by $\sum_{i=1}^N w_i\sigma_i$. For the second question, a similar argument based on convexity can be used.

5. This is done as in question 1 of Exercice 4 of TD5 with a small modification due to the presence of non-zero interest rate.

6. We start with the definition of the implied volatility:

$$C^{BS}(T, K, I(T, K)) = \sum_{i=1}^{N} w_i C^i(T, K)$$

Applying the put-call duality formula of the previous question on both sides yields

$$P^{BS}(T, \frac{S_0^2 e^{2rT}}{K}, I(T, K)) = \sum_{i=1}^{N} w_i P^i(T, \frac{S_0^2 e^{2rT}}{K})$$

Now applying the put-call parity to both sides, since the weights sum up to one, we get

$$C^{BS}(T, \frac{S_0^2 e^{2rT}}{K}, I(T, K)) = \sum_{i=1}^{N} w_i C^i(T, \frac{S_0^2 e^{2rT}}{K}),$$

which shows that the implied volatility for strike $\frac{S_0^2 e^{rT}}{K}$ coincides with the one for strike K.